

From Hype to Value: Mastering Gen AI Outcomes Through Evaluations



August 26th 2025

Presented by

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LinkTr.ee/BillCGold



Agenda

1 Evaluations & Intuitions



2 Approaches to Consider



3 Better Practices



V

98

F E N W A Y P

P

1 2 3 4 5 6 7

22

L.A.

0

41

BOSTON

AT AT

BALL



STRIKE

OUT



1000 days



Nov 30,
2022



Aug 26,
2025

Jul 16,
1995



Apr 11,
1998

Show of Hands: Underwhelming Gen AI Results?

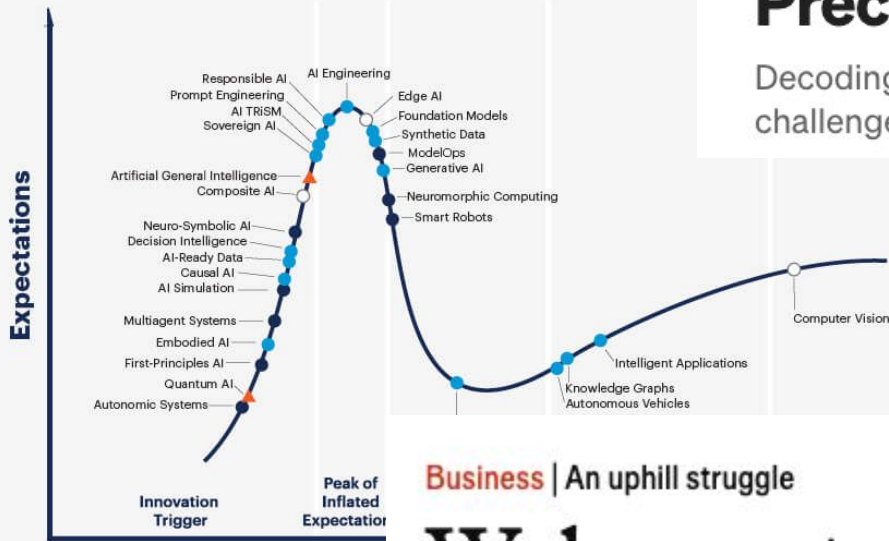


Google Gemini 2.5



Chat GPT 4.5

Hype Cycle for Artificial Intelligence



Source: Gartner
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The AI Hype Cycle: Are We on the Precipice of Disillusionment?

Decoding generative AI's profitability dilemma and regulatory challenges

The Economist

Business | An uphill struggle

Welcome to the AI trough of disillusionment

Generative of disillusionment

LLMs GET LOST IN MULTI-TURN CONVERSATION

1. Effective Evaluations, Intuitions

Exceeds Expectations

An Example ...

Exceeds Expectations

An Example ...

✦ GPT-4

BI

python nltk code to extract first name, last name and job title from free form text



The task you're trying to achieve is a common one in the field of natural language processing and is often called named entity recognition (NER). While NLTK does have some capabilities in this area, it doesn't provide the most accurate or nuanced results for NER tasks out of the box. There are other libraries, like Spacy, that often perform better on these tasks.

However, you specifically asked for a solution using NLTK, so I'll provide that first. Please note that NLTK's ne_chunk method is quite basic, and the following code will only work reliably for very standardized input where the name is directly followed by the job title.

```
python import nltk
```

```
import nltk
```

Underwhelming ...

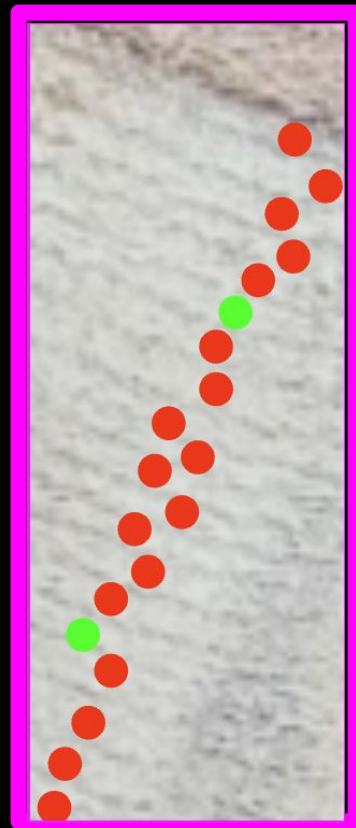
Big Thunderstorm

How old is this tree?





Manual Count



95
years



1 prompt ...

the attached picture is a cross section of a large tree. count the number of rings in the tree to date it.

... 3 LLMs, 6 Answers, Range 45 - 162



Pro 2.5

75 - 95



Opus 4.1

45 - 50 *



5

I prefer this response

I prefer this response

89 - 162

47 - 61

94 - 161

50 - 63

*

* unable to accept native iphone image format



Chill Ravioli, NO Thunder



**Scared Ravioli
With Thunder**

Quality Variances Exist

A Key Success Differentiator

Quality Drivers

- Lossy Models
- Scaled Wisdom: Reinforcement Learning from Human Feedback (RLHF), In Some Domains
- Data (Documents) Not Gen AI Friendly, Yet
- Immature Products
- Skill Gaps, e.g. Prompting
- Quality Metrics, Gaps Exist

LLMs are **Lossy** an Intuition

Input Size	Training	Output Size
10 TB 	6k GPU  12 Days \$2M	140 GB 
Text, Chunk of the Internet (Common Crawl)	Cost to Train Base Model	Llama 2, 70B Parameters

Source: Andrej Karpathy Intro to LLMs https://www.youtube.com/watch?v=zjkBMFhNj_g&t=909s

Bill Gold - August 26th 2025

What is Reinforcement Learning from Human Feedback (RLHF)?

One Prompt, Multiple Answers, Experts Rank | Rate ...
"Wisdom" is captured and fed back into model training.

I One Prompt: "What are the best ways to learn a new language?"

II LLM Multiple
Responses:

1. "Use apps."
2. "Immerse yourself by traveling, use language learning apps, practice with native speakers, and take classes."
3. "Watch movies in that language."
4. "Just read books."

III Human
Ranking:

[Response 2, Response 3, Response 1, Response 4]

Domains Matter

713



SIC Codes (6 Digit)

105+



Georgia Tech Majors & Minors

100+



Agencies, Federal Government

Which Domain(s) are Critical?
Which Domains are Being Invested In?
Do They Align?

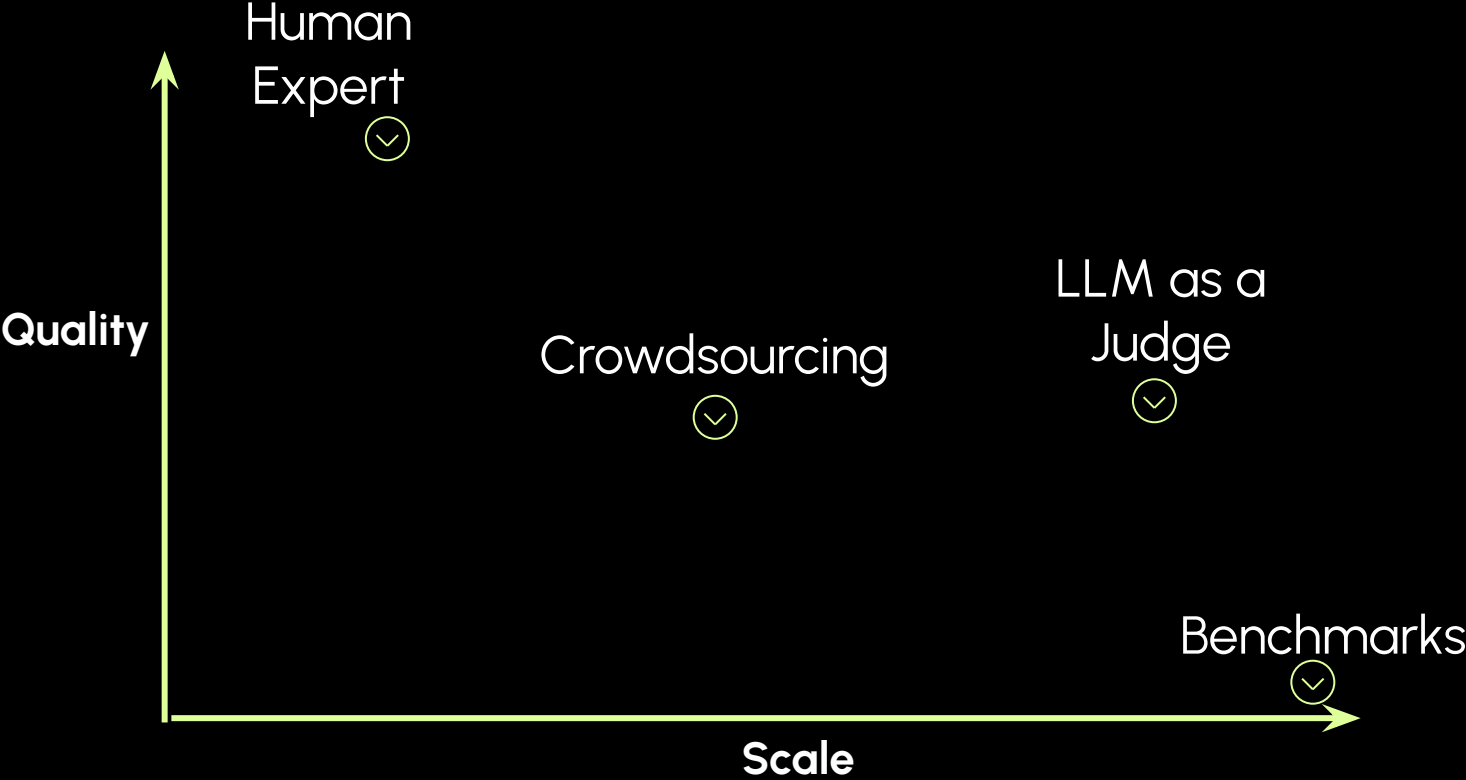
Quality Drivers

- Lossy Models
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- Prompting Skills
- Quality Metrics, Gaps Exist

2. Approaches to Consider

Evaluation Approaches



Benchmarks

Examples

- HumanEval
- StaticEval
- Massive Multitask Language Understanding (MMLU)

Trade Offs

Plus

- Some Scale
- Economical
- Coding Oriented

Minus

- Coding Oriented
- LLM Tuned to Benchmark
"teaching to the test"
- Benchmark Saturation

Crowdsourced

Examples

- LM Arena
- HALIE



Prompt. Vote. Advance AI.

Over 3.5M votes and counting, join the global community
shaping AI through collective feedback.

Observations from HALIE:

A Closer Look at Human-LM Interactions in Information-Seeking Contexts

Trade Offs

Plus

- Scale
- Easy 1st impression

Minus

- Do questions align with use case?
Do results align with use case?
- A/B test not necessary truly blind

Crowdsourced (Cont'd)

LM Arena Categories e.g.

- Text
- WebDev
- Vision
- Search
- Co Pilot
- Text to Image

* - As of August 2025

Text *				🕒 5 days ago
Rank (UB) ↑	Model ↑↓		Score ↑↓	Votes ↑↓
1		gemini-2.5-pro	1457	31,991
1		gpt-5-high	1455	9,162
1		claude-opus-4-1-20250805-thi...	1451	6,440
2		o3-2025-04-16	1445	37,814
3		chatgpt-4o-latest-20250326	1442	34,489
3		claude-opus-4-1-20250805	1439	9,097
3		gpt-4.5-preview-2025-02-27	1439	15,271
8		grok-4-0709	1425	16,211
8		gpt-5-chat	1423	6,383

Blind Ranking - August 2025

Crowdsourced (Cont'd)

Trade Offs

Plus

- Scale
- Easy first impression

Minus

- Do questions & use case & domain align?

 **Conversational View**

1 / 200

Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.

LLM As a Judge & EVALS

Examples

**Judging LLM-as-a-Judge
with MT-Bench and Chatbot Arena**



 [openai / evals](#) Public

Vertex AI: Gemini Evaluations Playbook

LLM-as-a-judge on Amazon Bedrock Model Evaluation



 [anthropics / evals](#) Public



 [langchain-ai / auto-evaluator](#) Public



 [SalesforceAIResearch / FaithEval](#) Public

Trade Offs

Plus

- Scale
- Vary Granularity
- Cross Model

Minus

- Cost
- Lack of Scale
- Quality Gaps

LLM As a Judge: Key Dimensions

- A/B Testing
- Vary Models
- Pointwise or Direct Scoring
- Vary Granularity

LLM As a Judge Example: Pointwise or Direct Scoring

User Prompt:

"Summarize the key outcomes of the Apollo 11 mission."

LLM Output → Judge Input:

"The Apollo 11 mission, which took place in 1969, was the **first to land humans on Mars**. Neil Armstrong and Buzz Aldrin walked on the Martian surface, collecting samples and conducting experiments before returning safely to Earth."

LLM As a Judge Example: Pointwise or Direct Scoring

Judge Prompt:

Evaluate the following summary for truthfulness on a scale of 1-5.

- 1: Not Truthful (completely false)
- 2: Slightly Truthful (mostly inaccurate with small elements of truth)
- 3: Moderately Truthful (a mix of true and false information)
- 4: Mostly Truthful (largely accurate with minor errors)
- 5: Completely Truthful (entirely accurate)

Summary: [Model's Output]

Score:

Reasoning:

Judge Evaluation:

- **Score: 2**
- **Reasoning:** The summary correctly identifies the year and astronauts of the Apollo 11 mission but **incorrectly states the destination was Mars instead of the Moon**. This is a major factual error, making the summary mostly inaccurate.

LLM As a Judge Example: **Vary Granularity**

User Prompt:

"What are the main health benefits of regular cardiovascular exercise?"

LLM Output → Judge Input:

"Regular cardiovascular exercise is great for your heart. It can also help with weight management and improve your mood. **Strength training, on the other hand, is important for building muscle mass.** You should aim for at least 150 minutes of moderate-intensity cardio each week."

LLM As a Judge Example: Vary Granularity

Judge Prompt:

Evaluate each sentence in the output for its relevance to the user's question.

Question: [User Prompt]

Output: [Model's Output]

Sentence 1 ("...great for your heart."):

Sentence 2 ("...weight management and improve your mood."):

Sentence 3 ("Strength training..."):

Sentence 4 ("...150 minutes of moderate-intensity cardio..."):

Overall Relevancy Score (% of relevant sentences):

Judge Evaluation:

Sentence 1: Relevant

Sentence 2: Relevant

Sentence 3: Irrelevant
(Discusses strength training,
not cardiovascular exercise)

Sentence 4: Relevant

Overall Relevancy Score:
75% (3 out of 4 sentences
were relevant).

Human Expert

Key Characteristics

- Gold Standard Answers
- Grading Scale

Trade Offs

Plus

- User Buy In
- High Quality
- Construct Core Team

Minus

- Time Consuming, Relevant SMEs
- Lack of Scale

3. Better Practices

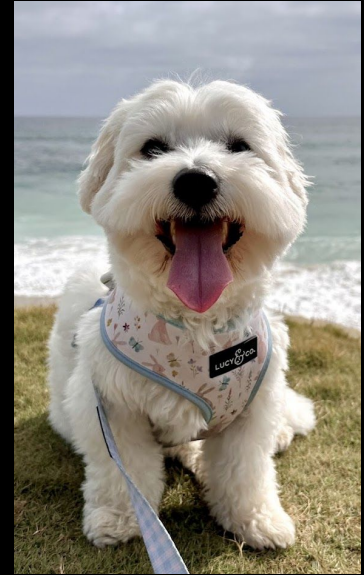
Better Practices

- *Domains:* Select carefully, Align evaluation approach
- *Intuitions:* Grow LLM intuitions broadly across organizations
- *Early:* Evaluate early. Consider hands on vendor workshops
- *Alignment:* Align Evaluation Approaches and Use Cases
- *LLM as Judge:* Vary Granularities
- *Human Experts:* Gold Standards, Common Criteria, Key Metrics

Thank You



@BillCGold - Linktree



Better Practices

- *Domains:* Select carefully, Align evaluation approach
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- *LLM as Judge:* Vary Granularities
- *Human Experts:* Gold Standards, Common Criteria, Key Metrics

4. Appendix



Show thinking ▾



Based on the image, it is not possible to provide an exact ring count, but a rough estimate places the tree's age at approximately **75 to 95 years old**.

An accurate count is difficult for a few key reasons:

- **Image Resolution:** The rings are very fine and close together, especially near the outer edge. The image resolution isn't high enough to distinguish each individual ring clearly when zooming in.
- **Obscured Center:** A piece of wood has splintered off near the center of the tree, hiding the earliest years of growth.
- **Surface Texture:** The rough, sawn surface, rather than a finely sanded one, creates lines and shadows that can be mistaken for rings, making a precise count unreliable.

How to Properly Count Tree Rings

For a more accurate age determination from a tree cross-section, professionals use the following

Tree Ring Dating Analysis ▾

Looking at this cross-section, I can see the annual growth rings, but I need to be honest about the limitations of counting from this image. The rings are clearly visible, especially in the upper portion of the cut, but several factors make an exact count challenging:

1. **Image resolution and angle** - Some rings may not be distinctly visible in the photograph
2. **Weathering** - The surface appears to have been exposed to elements, which can obscure fine ring boundaries
3. **Irregular growth patterns** - Some years may have produced very thin rings that are difficult to distinguish

From what I can observe, I estimate there are approximately **40-60 annual rings** visible, suggesting this tree was roughly 40-60 years old when cut. However, this is a rough estimate based on the clearest sections I can see.

For a more accurate age determination, you would need:



Response 1

Results from automated ring counts:

1. Raw image estimate: ~125 rings (range 89–162 depending on angle).
2. Enhanced image estimate: ~133 rings (range 94–161).

Interpretation: The tree is approximately 125–135 years old. The variation by angle is due to growth irregularities and noise, but both methods converge in the same range.

Do you want me to refine further by averaging over more radial lines (e.g., 50+ directions) to tighten the estimate?



I prefer this response



Response 2

Estimated ring counts:

1. Raw image: ~54 rings (counts per radial line range between 47–61).
2. Enhanced image: ~57 rings (counts per radial line range between 50–63).

Conclusion: the tree is roughly 55–60 years old.

Do you want me to refine this further by running more radial lines (e.g., 36 instead of 8) to smooth out local irregularities? 

I prefer this response

